

Link discord IOAI untuk diskusi, gathering, dan having fun: <https://discord.gg/CSvBqhR3>

Sesi Daring #1 (5 hari)						
No	Hari & Tanggal	Sesi	Materi	Target Capaian Pembelajaran Minimal	Link Sesi Online	Pemateri
1	Minggu, 8 Maret 2026	Siang (13.00-15.00)	Python: basic & advanced; Pandas (Data Handling); Python libraries: numpy & torch (struktur data) Slide: Link Recording: Link Passcode: d^5NUmf* Homework: Link Deadline: Minggu, 15 Maret 2026 pkl 23:59 WIB	Implementasi pipeline data sederhana (load-clean-transform), data handling, operasi vektorisasi menggunakan numpy dan pytorch,	https://ui-ac-id.zoom.us/j/93792060660?pwd=bFkT87OafMjpb8VA6lVrLaYZCBd8BT.1	Faiz Rizki (IA-TOKI, Silver Medalist IOAI 2025)
		Sore (16.00-18.00)	Sesi Latihan Kaggle: Link Deadline: Minggu, 8 Maret 2026 pkl 18:00 WIB			Faiz Rizki (IA-TOKI, Silver Medalist IOAI 2025)
2	Selasa, 10 Maret 2026	Siang (13.00-15.00)	Probability: Bayesian Theorem, Naive Bayes (Refresh: Basic of Probability) Slide: Link Recording: Link Passcode: LY.4YD7W	Mampu menghitung dan membedakan probabilitas, likelihood, posterior, dan prior. Asumsi Naive Bayes dan penggunaannya (coding), dari probabilitas ke decision.		Dr. Dina Chahyati (Fasilkom UI)
		Sore (16.00-18.00)	Statistics: distribusi (uniform, normal, binomial, multinomial, poisson), correlation & causation, (Refresh: Mean, Median, Modus, Standard	Diketahui fenomena dan data, pilih distribusi yang sesuai. Korelasi vs kausalitas.		Dr.-Ing. Ikhsanul Habibie (Fasilkom UI)

			Deviasi, Variance) Slide: Recording: Link Passcode: ac!Dij9r			
			Sesi Latihan Kaggle: Link Deadline: Selasa, 10 Maret 2026 pk1 22:00 WIB			
3	Rabu, 11 Maret 2026	Siang (13.00-15.00)	Data Cleaning, Feature Scaling, Normalization (z-score, min-max) Slide: Link Recording: Link Passcode: bZ@M@6!R	Mampu melakukan normalisasi dan scaling, melakukan visualisasi data untuk menemukan permasalahan pada data (outlier, skewness, imbalance).		Dr.-Ing. Ikhsanul Habibie (Fasilkom UI)
		Sore (16.00-18.00)	Sesi Latihan Kaggle: Link Deadline: Rabu, 11 Maret 2026 pk1 22:00 WIB			Dr.-Ing. Ikhsanul Habibie (Fasilkom UI)
4	Kamis, 12 Maret 2026	Siang (13.00-15.00)	Concept of AI/ML, Linear Model, Bias & Variance Tradeoff Slide: Link Recording: Link Passcode: Gr#1Ei#6	Mengerti paradigma ML yg ada dan kapan digunakan. Trade-off bias vs. variance.		M. Febrian Rachmadi, Ph.D. (Fasilkom UI)
		Sore (16.00-18.00)	K-NN and K-Means Slide: Link Recording: Link Passcode: BHP@QE=9	Fungsi objektif regresi dan maknanya. Beberapa model ML baik konsep maupun kode.		M. Febrian Rachmadi, Ph.D. (Fasilkom UI)

			Sesi Latihan Kaggle #1: Link Kaggle #2: Link Deadline: Kamis, 12 Maret 2026 pkl 22:00 WIB			
5	Jum'at, 13 Maret 2026	Siang (13.00-15.30)	Skema hold-out, cross-validation, underfitting, overfitting Slide: Link Recording: Link Passcode: @7Ft?A@h	Penggunaan metrik evaluasi dan implikasi pada keputusan. Mendeteksi overfitting/underfitting dari hasil eksperimen. Menjelaskan dan mengidentifikasi permasalahan yg dialami model secara sistematis.		Dr. Masayu Leylia Khodra (STEI ITB)
		Sore (15.45-17.30)	Evaluation Metrics (Accuracy, F1, precision, recall, MSE, R2), Confusion Metrics Slide: Link Recording: Link Passcode: @7Ft?A@h			Dr. Masayu Leylia Khodra (STEI ITB)
			Kaggle: Link Deadline: Jum'at, 13 Maret 2026 pkl 22:00 WIB			

Sesi Daring #2 (5 hari)						
No	Hari & Tanggal	Sesi	Materi	Target Capaian Pembelajaran Minimal	Link Sesi Online	Pemateri
6	Minggu, 29 Maret 2026	Siang (13.00-15.00)	Decision Trees; Regression Trees, Basic Ensemble & Random Forest	Memahami konsep impurity, split, dan overfitting pada tree. Memahami mengapa ensemble bekerja (variance	https://ui-ac-id.zoom.us/	Faiz Rizki (IA-TOKI, IOAI 2025)

			Slide: Link Recording: Link Passcode: Kp&APM.5	reduction).	j/93792060660?pwd=bFkT87OafMjpb8VA6lVrLaYZCB D8BT.1	
		Sore (16.00-18.00)	Kaggle: Link Pengumpulan: Link Deadline: Minggu, 29 Maret 2026 pkl 22:00 WIB			Faiz Rizki (IA-TOKI, IOAI 2025)
7	Senin, 30 Maret 2026	Siang (13.30-15.00)	Support Vector Machine (SVM): binary classification & Support Vector Regression SVM: multiclass classification; Python library: SVC Slide: Link Recording: Link Passcode: \$p2&f6p4	Memahami margin, hyperplane secara geometris, kernel trick, peran parameter C, binary vs multiclass SVM, klasifikasi SVM vs regresi SVR.		Belati Jagad Bintang Syuhada
		Sore (16.00-18.00)	Google Colab: Link			Belati Jagad Bintang Syuhada
8	Selasa, 31 Maret 2026	Siang (13.00-15.00)	Perceptron, MLP, fully connected layer (various architectures) Slide: Link Recording: Link Passcode: mNi+eU^1	Memahami neuron sbg fungsi matematika, forward vs backward propagation, non-linearity pada activation function, loss function, basic gradient descent.		Dr. Eng. Laksmita Rahadiani (Fasilkom UI)
		Sore (16.00-18.00)	ANN Basics: struktur jaringan, forward propagation. Activation Functions (ReLU, Sigmoid, Tanh) Slide: Link Recording: Link Passcode: +9W\$6!uM			Dr. Eng. Laksmita Rahadiani (Fasilkom UI)

			Kaggle: Link Forn: Link Onerender: Link			
9	Rabu, 1 April 2026	Siang (13.00-15.00)	Gradient Descent, Backpropagation, Automatic Differentiation (autodiff), NN loss function Slide: Link Recording: Link Passcode: *&61fNtt	Memahami bagaimana NN bisa dioptimasi menggunakan backpropagation, autodiff, dan berbagai loss functions. Memahami curse of dimensionality, feature engineering, dan dim reduction.		Rifki Afina Putri, Ph.D. (Ilkom UGM)
		Sore (16.00-18.00)	Feature Engineering and Dimensionality Reduction (e.g., PCA, t-SNE, UMAP, etc.) Slide: Link Recording: Link Passcode: W!Tw601h Kaggle: Link			Rifki Afina Putri, Ph.D. (Ilkom UGM)
10	Kamis, 2 April 2026	Siang (13.00-15.00)	Ensemble methods; cross-validation for ensemble inference Random Forests: sklearn.ensemble.RandomForestClassifier Slide: Link Recording: Link Passcode: G*5ngafv	Penjelasan lebih dalam dari ensemble learning, contoh kasus Random Forest ataupun ensemble lainnya, termasuk cross-validation. Contoh implementasi & latihan.		Jaycent Gunawan Ongris, S.Kom. (Postgrad student at Radboud University)
		Sore (16.00-18.00)				Jaycent Gunawan Ongris, S.Kom. (Postgrad student at Radboud University)

Luring di Fasilkom UI (14 hari)						
No	Hari & Tanggal	Sesi	Materi	Tempat	Pemateri	
1	Senin, 6 April 2026	Pagi (08.00-11.30)	Kedatangan peserta di Hotel	Hotel		
		Siang (12.30-16.00)	Kedatangan peserta di Hotel	Hotel	M. Febrian Rachmadi, Ph.D.	
		Sore (16.00-17.30)	Briefing 1 (Sore)	Hotel	M. Febrian Rachmadi, Ph.D.; Alfan Farizki, Ph.D.	
		Malam (19.00-21.00)	Briefing 2 (Malam)	Hotel	M. Febrian Rachmadi, Ph.D., Dr. Mushthofa	
2	Selasa, 7 April 2026	Pagi (08.00-08.30)	Pembukaan Resmi di Fasilkom UI	Ruang Sidang Lt. 4		
		Pagi (08.30-11.30)	[TKD] Linear Models & Gradient Descent	Lab A3.02	Adila Alfa Krisnadhi, Ph.D	
		Siang (12.30-14.30)	[TKD] Neural Networks, Backpropagation, NN loss functions	Lab A3.02	Adila Alfa Krisnadhi, Ph.D	
		Sore (14.30-17.30)	[TKD] Optimization (Adam family, after loss function), loss functions	Lab A3.02	Adila Alfa Krisnadhi, Ph.D	
		Malam (19.00-21.00)	[UPSOLVE] Upsolving	Lab A3.02		
3	Rabu, 8 April 2026	Pagi (08.00-11.30)	[PROG] Pre-simulation	A2.06 (Laptop)	Nyoo Steven, S.Kom.	
		Siang (12.30-14.30)	[TKD] Optimization (Adam family, after loss function), loss functions	A2.06 (Laptop)	Adila Alfa Krisnadhi, Ph.D	
		Sore (14.30-17.30)	[PROG] Praktik kasus dan Latihan Soal	A2.06 (Laptop)	Adila Alfa Krisnadhi, Ph.D	
		Malam (19.00-21.00)	[UPSOLVE] Upsolving	A2.06 (Laptop)		
4	Kamis, 9 April 2026	Pagi (08.00-11.30)	[TKD] Gradient Boosting (Catboost, XGboost, LGBM)	A2.06 (Laptop)	Nyoo Steven, S.Kom.	
		Siang (12.30-14.30)	[TKD] Praktik kasus	A2.06	Nyoo Steven, S.Kom.	

				(Laptop)	
		Sore (14.30-17.30)	[PROG] Latihan Soal	A2.06 (Laptop)	Darren
		Malam (19.00-21.00)	[UPSOLVE] <i>Upsolving</i>	A2.06 (Laptop)	
5	Jumat, 10 April 2026	Pagi (08.30-11.30)	[TKD] Feature Engineering: Principal Component Analysis (PCA); Dimensionality Reduction Methods: t-SNE, UMAP, Other Dimensionality Reduction Methods; Data Augmentation and Startification	A2.06 (Laptop)	Dr. Eng. Laksmita Rahadiani (Fasilkom UI)
		Siang (12.30-14.30)	[PROG] Latihan Soal	A2.06 (Laptop)	Nyoo Steven, S.Kom.
		Sore (14.30-17.30)	[PROG] Latihan Soal	A2.06 (Laptop)	Nyoo Steven, S.Kom.
		Malam (19.00-21.00)	[UPSOLVE] <i>Upsolving</i>	A2.06 (Laptop)	
6	Sabtu, 11 April 2026	Pagi (08.00-11.30)	[SIM] Simulasi	A2.06 (Laptop)	Rifki Afina Putri, Ph.D., Dr. Mushthofa
		Siang (12.30-14.30)	[SIM] Simulasi	A2.06 (Laptop)	Rifki Afina Putri, Ph.D., Dr. Mushthofa
		Sore (14.30-17.30)	[SIM] Simulasi	A2.06 (Laptop)	Rifki Afina Putri, Ph.D., Dr. Mushthofa
		Malam (19.00-21.00)	[UPSOLVE] <i>Upsolving</i>	A2.06 (Laptop)	Rifki Afina Putri, Ph.D., Dr. Mushthofa
7	Minggu, 12 April 2026		Break		
8	Senin, 13 April 2026	Pagi (08.00-11.30)	Hierarchical Clustering	A3.02	Dr. Mushthofa
		Siang (12.30-14.30)	Praktik kasus	A3.02	Dr. Mushthofa
		Sore (14.30-17.30)	[PROG] Latihan Soal	A3.02	Dr. Mushthofa, Nyoo Steven, S.Kom.

		Malam (19.00-21.00)	[UPSOLVE] <i>Upsolving</i>	A3.02	
9	Selasa, 14 April 2026	Pagi (08.00-09.50)	[TKD] Convolutional Layers (1D), Recurrent Neural Network, Bidirectional RNN	A3.02	Dr.-Ing. Ikhsanul Habibie
		Pagi (10.00-12.00)	[PROG] Latihan Soal	A3.02	Belati, Eryawan
		Siang (13.00-15.00)	[TKD] Convolutional Layers (1D), Recurrent Neural Network, Bidirectional RNN	A3.02	Dr.-Ing. Ikhsanul Habibie
		Sore (15.30-17.30)	[PROG] Latihan Soal	A3.02	Belati, Eryawan
		Malam (19.00-21.00)	[UPSOLVE] <i>Upsolving</i>	A3.02	
10	Rabu, 15 April 2026	Pagi (08.00-11.30)	[TKD, PROG] Learning techniques for ANN: Momentum Methods (Adam, AdamW) Adaptive Learning Rates Convergence and Learning Rates Weight Regularization Early Stopping Dropout, Gaussian Noise Weight Initialization Batch Normalization;	A3.02	Rifki Afina Putri, Ph.D.
		Siang (12.30-14.30)	[TKD, PROG] Praktik kasus: ANN, CNN	A3.02	Rifki Afina Putri, Ph.D.
		Sore (14.30-17.30)	[PROG] Latihan Soal	A3.02	Rifki Afina Putri, Ph.D.
		Malam (19.00-21.00)	[UPSOLVE] <i>Upsolving</i>	A3.02	
11	Kamis, 16 April 2026	Pagi (08.00-11.30)	[TKD, PROG] Convolutional Layers (1D, 2D, 3D) [TKD, PROG] Convolutional Layers (Kernels, Convolution) [TKD, PROG] Pooling Techniques (Max, Average)	A7.13 (laptop)	M. Febrian Rachmadi, Ph.D.
		Siang (12.30-14.30)	[PROG] Praktik kasus: time-series, sequence	A7.13 (laptop)	M. Febrian Rachmadi, Ph.D.
		Sore (14.30-17.30)	[PROG] Latihan Soal	A7.13 (laptop)	M. Febrian Rachmadi, Ph.D.
		Malam (19.00-21.00)	[UPSOLVE] <i>Upsolving</i>	A7.13 (laptop)	

12	Jumat, 17 April 2026	Pagi (08.00-11.30)	[TKD, PROG] LSTM and variants (GRU, Bi-LSTM)	A7.13 (laptop)	Alfan Farizki Wicaksono, Ph.D.
		Siang (12.30-14.30)	[PROG] Praktik kasus: audio/text	A7.13 (laptop)	Alfan Farizki Wicaksono, Ph.D.
		Sore (14.30-17.30)	[PROG] Latihan Soal	A7.13 (laptop)	Alfan Farizki Wicaksono, Ph.D.
		Malam (19.00-21.00)	[UPSOLVE] <i>Upsolving</i>	A7.13 (laptop)	
13	Sabtu, 18 April 2026	Pagi (08.00-11.30)	[SIM] Simulasi	A3.02	Rifki Afina Putri, Ph.D., Dr. Mushthofa
		Siang (12.30-14.30)	[SIM] Simulasi	A3.02	Rifki Afina Putri, Ph.D., Dr. Mushthofa
		Sore (14.30-17.30)	[SIM] Simulasi	A3.02	Rifki Afina Putri, Ph.D., Dr. Mushthofa
14	Minggu, 19 April 2026	Pagi (08.00-11.30)	Pengumuman seleksi, persiapan kembali ke kota masing-masing	A3.02	